MATH 189 HW 5

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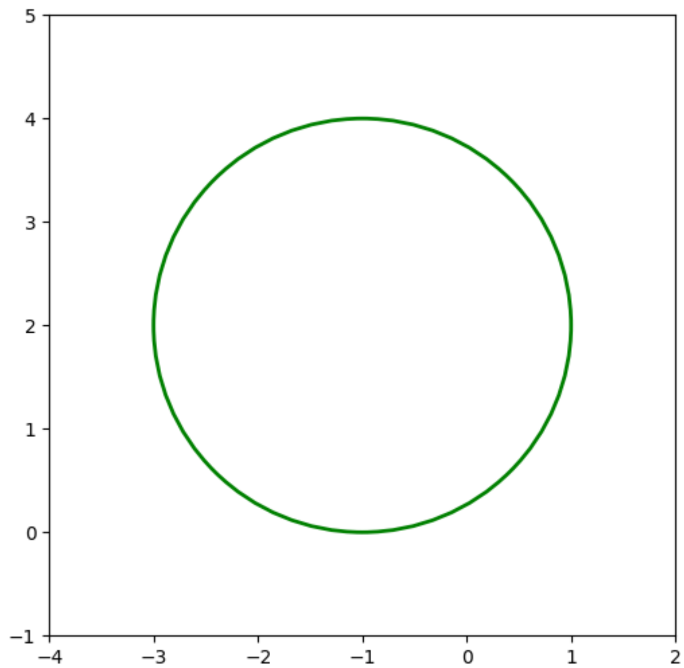
Question 1
Since boosting takes linear combinations of depth-one trees, which are univariate functions, we get the sum of univariate functions, which is just an additive model.

Question 2
We split each cell by finding the predictor X_j and cutpoint s that minimizes

$$\sum_{i:X_{ij}<s}(Y_i - \hat{Y}_{R_1})^2 + \sum_{i:X_{ij}\geq s}(Y_i - \hat{Y}_{R_2})^2$$

Question 3

a. 

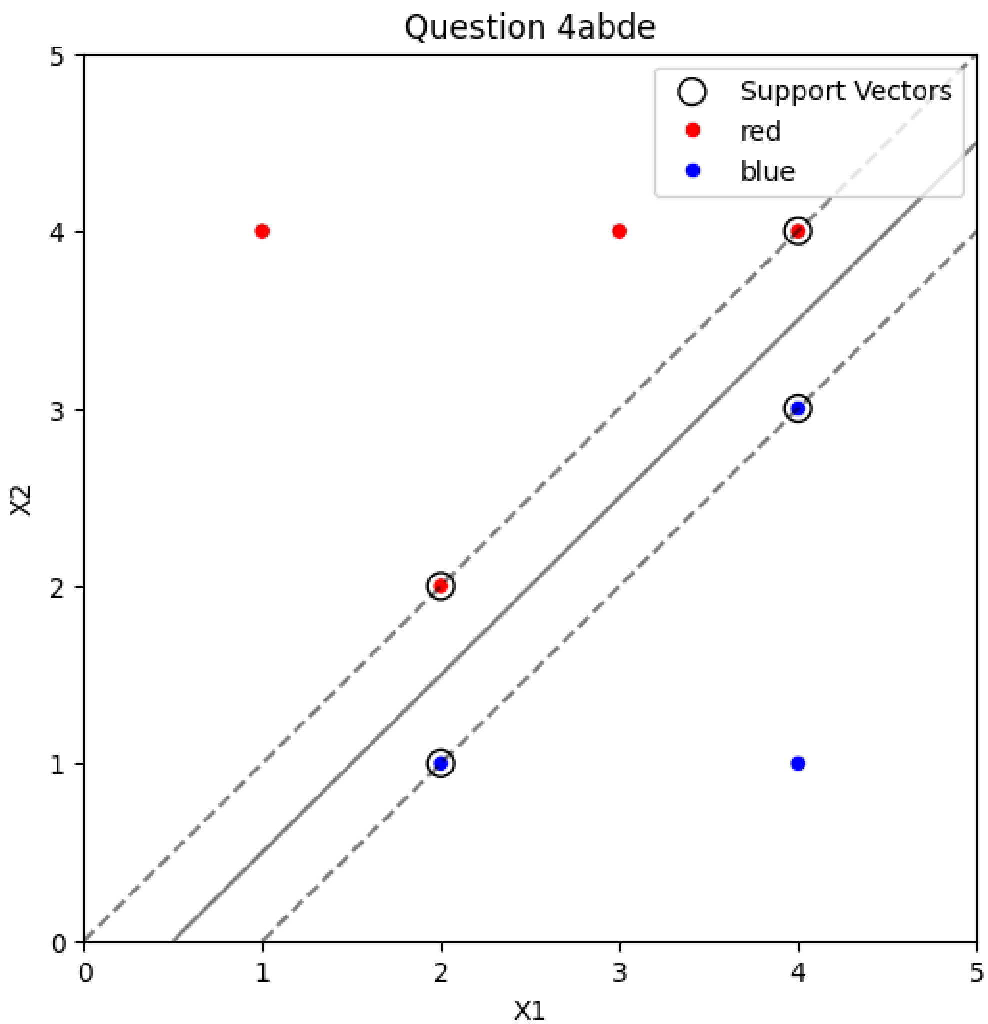
b. The set of points for which $(1+X_1)^2+(2-X_2)^2 > 4$ are the points outside the circle, whereas the set of points for which $(1+X_1)^2+(2-X_2)^2 \leq 4$ are the points on and inside the circle.

c.

Points	$(1+X_1)^2+(2-X_2)^2$	Class
(0,0)	5	Blue
(-1,1)	1	Red
(2,2)	9	Blue
(3,8)	52	Blue

d. If we expand the equation for the decision boundary to $5+2X_1-4X_2+X_1^2+X_2^2=4$, we get an equation that is linear in terms of X_1, X_2, X_1^2, X_2^2

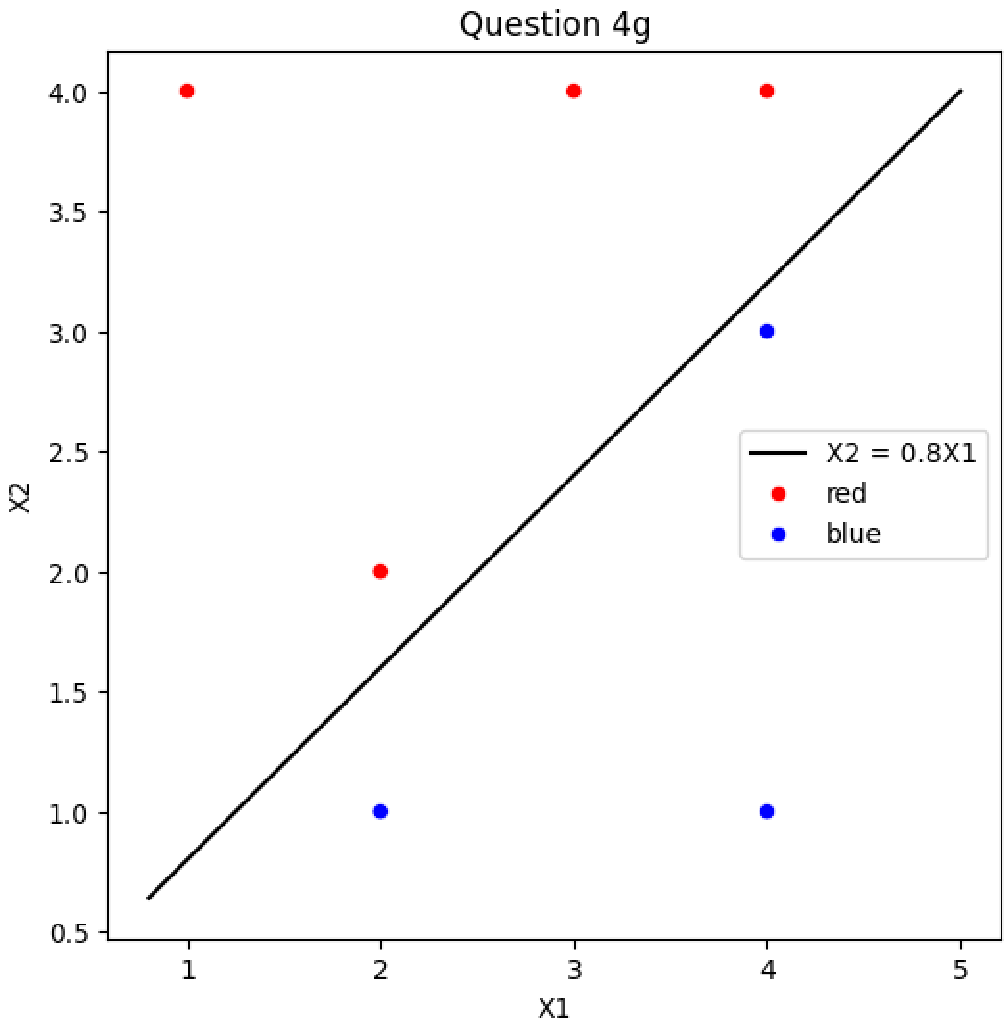
Question 4



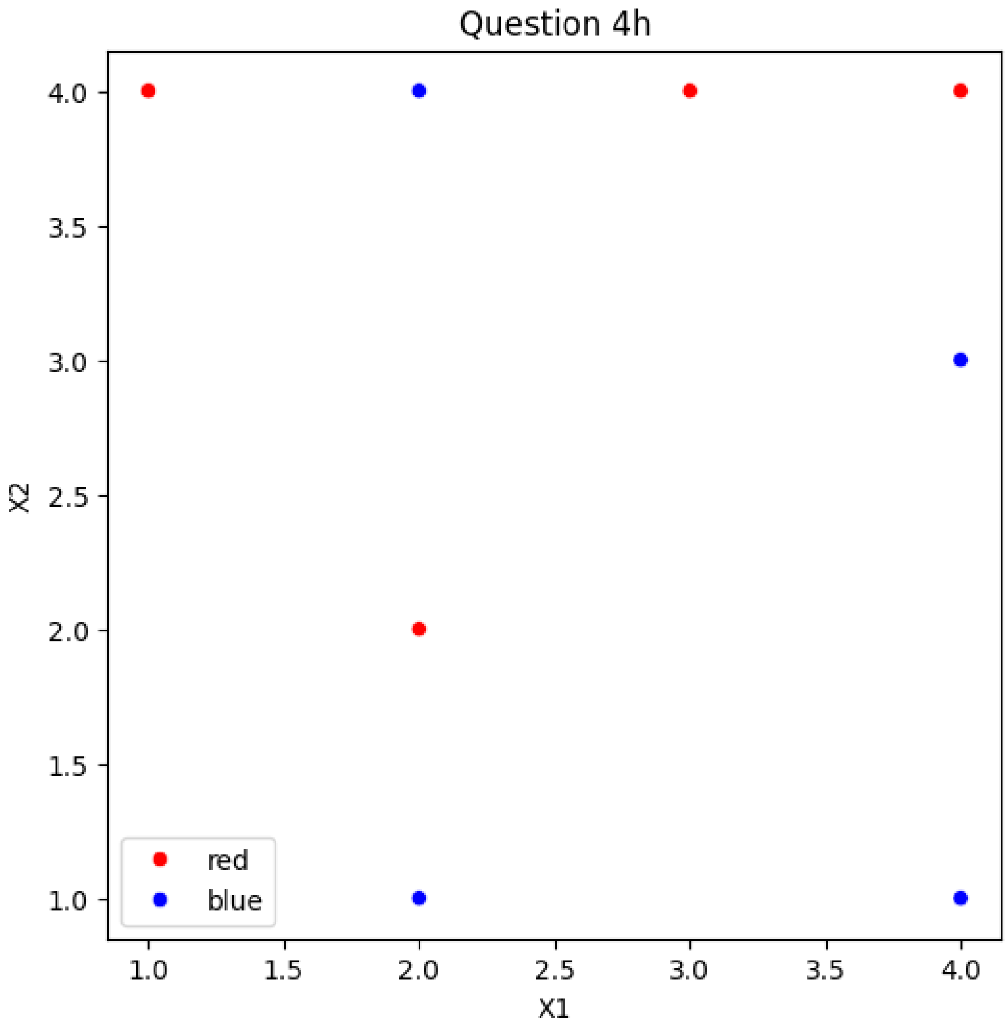
c. Classify to Red if $\beta_0 + \beta_1 X_1 + \beta_2 X_2 > 0$ and Blue otherwise.
In this case, $\beta_0 = -0.5,\ \beta_1 = 1,\ \beta_2 = -1$

f. The 7th observation (4,1) if moved slightly will not affect the hyperplane because it is not a support vector, the equation of the hyperplane does not depend on it.

g. The equation for this hyperplane is $0 = 0.8X_1 - X_2$

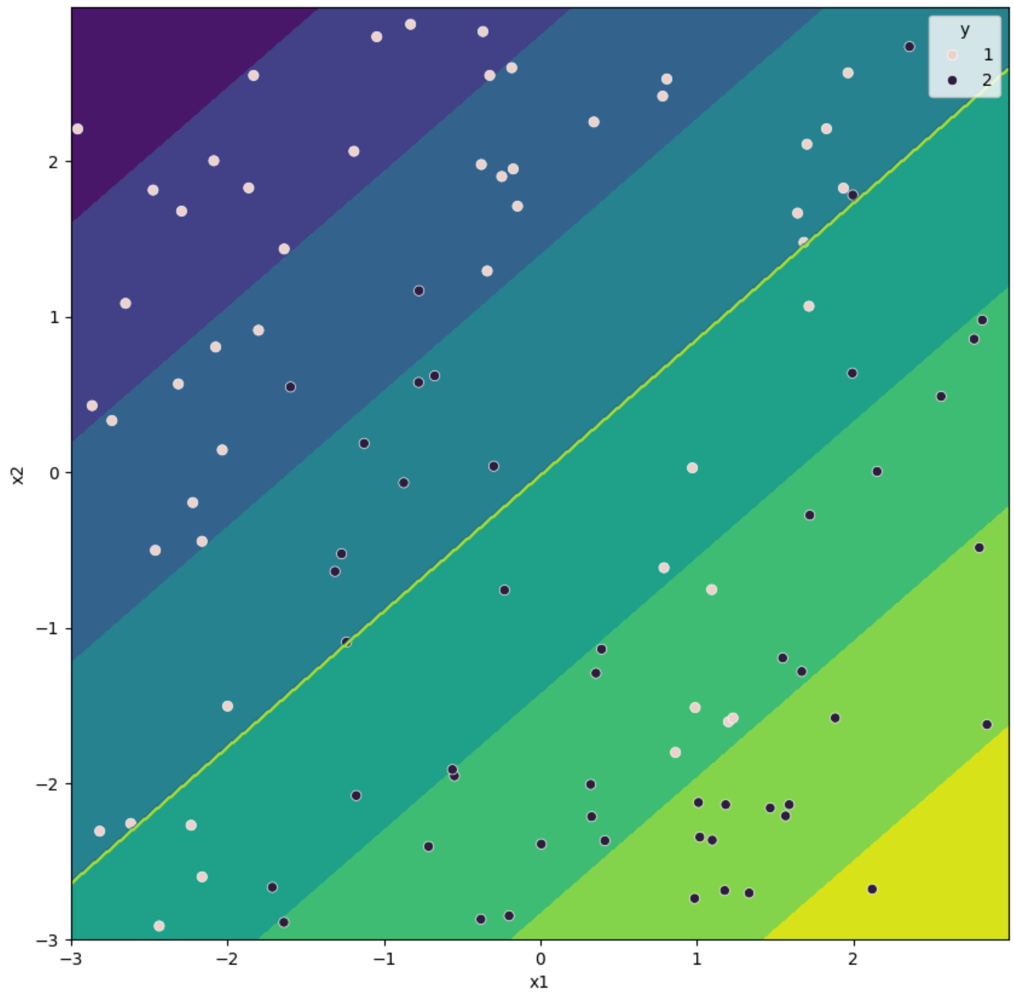

Question 4g
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
X2
1
2
3
4
5
X1
X2 = 0.8X1
red
blue

h. A point that would make the two classes unseparable by a hyperplane would be (2,4) with the category Blue.

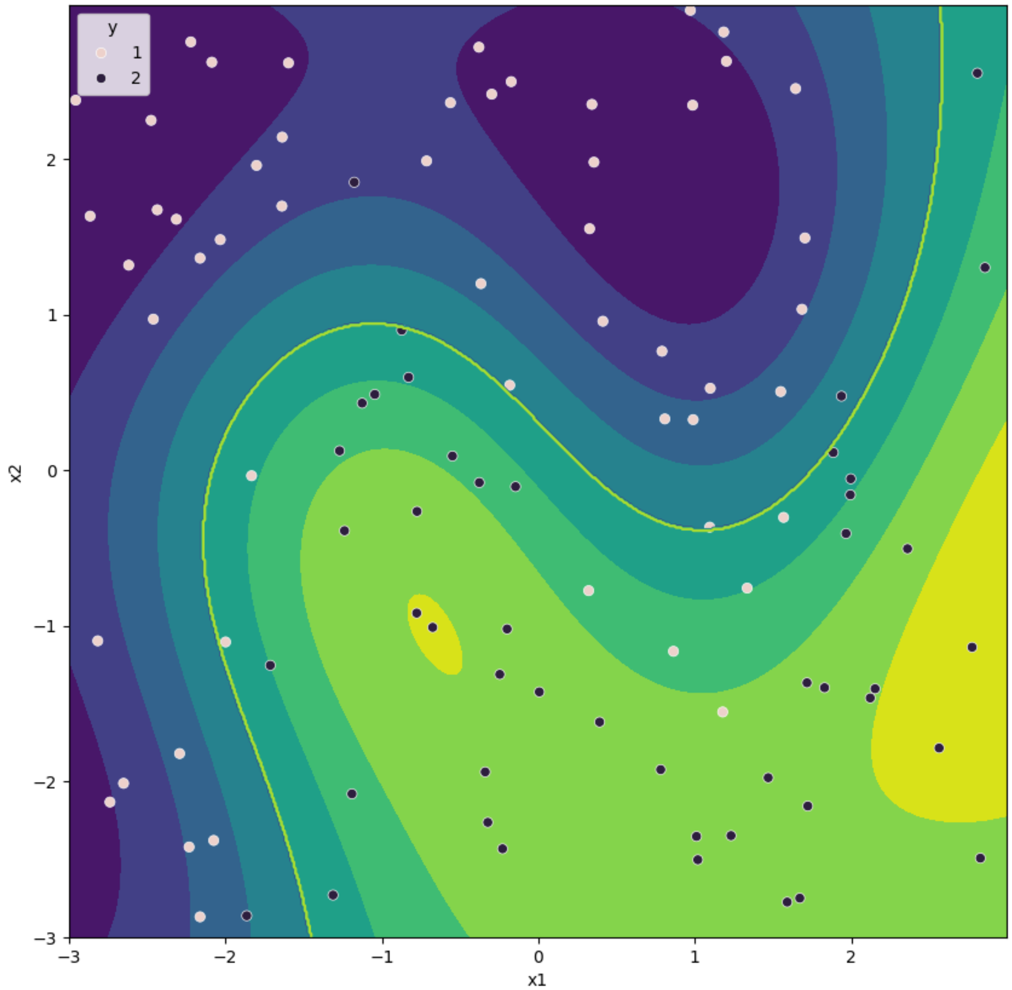


Question 5

Question 6
Linear kernel support vector machine
Training score: 0.77
Test score: 0.72



Radial kernel support vector machine
Training score: 0.89
Test score: 0.79



Therefore, radial kernel works better in this case of non-linear separation between the two classes.